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Lecture #5: Backdoor Defense

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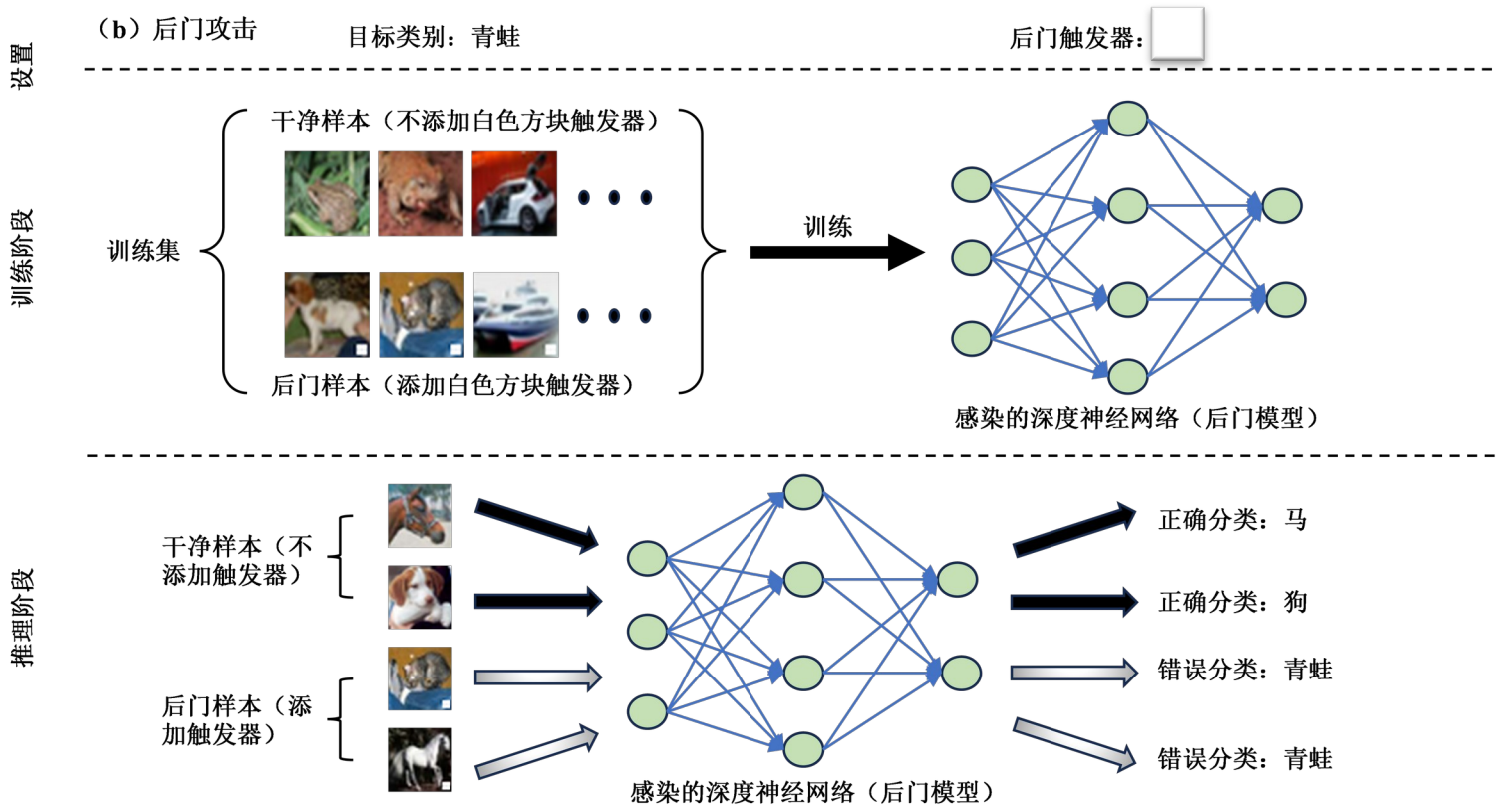
Outline

- Overview
- Backdoor defense in AI
- Backdoor defense taxonomy
- Anti-Backdoor Learning
- Neural Cleanse

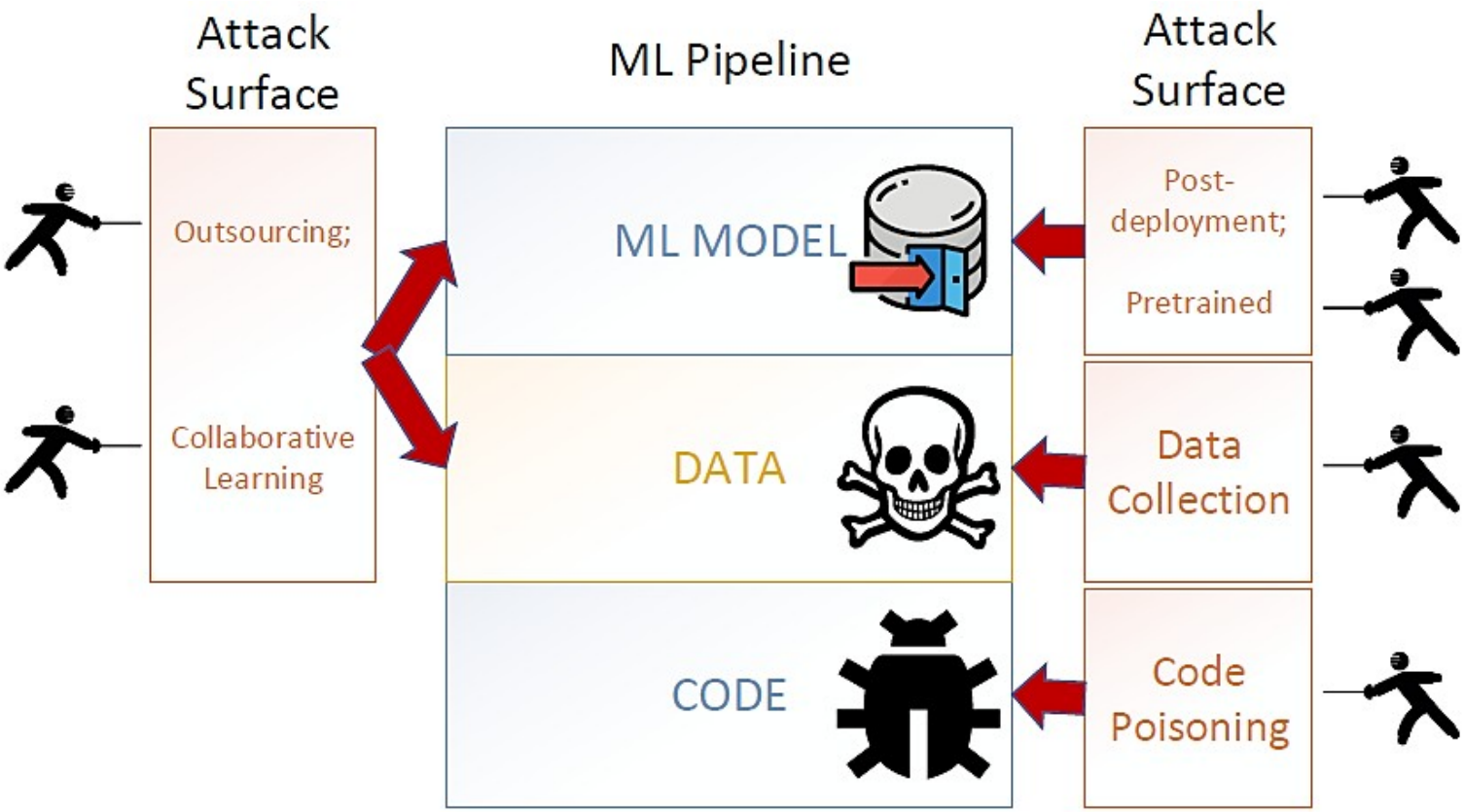
Overview

- Backdoor Attack:



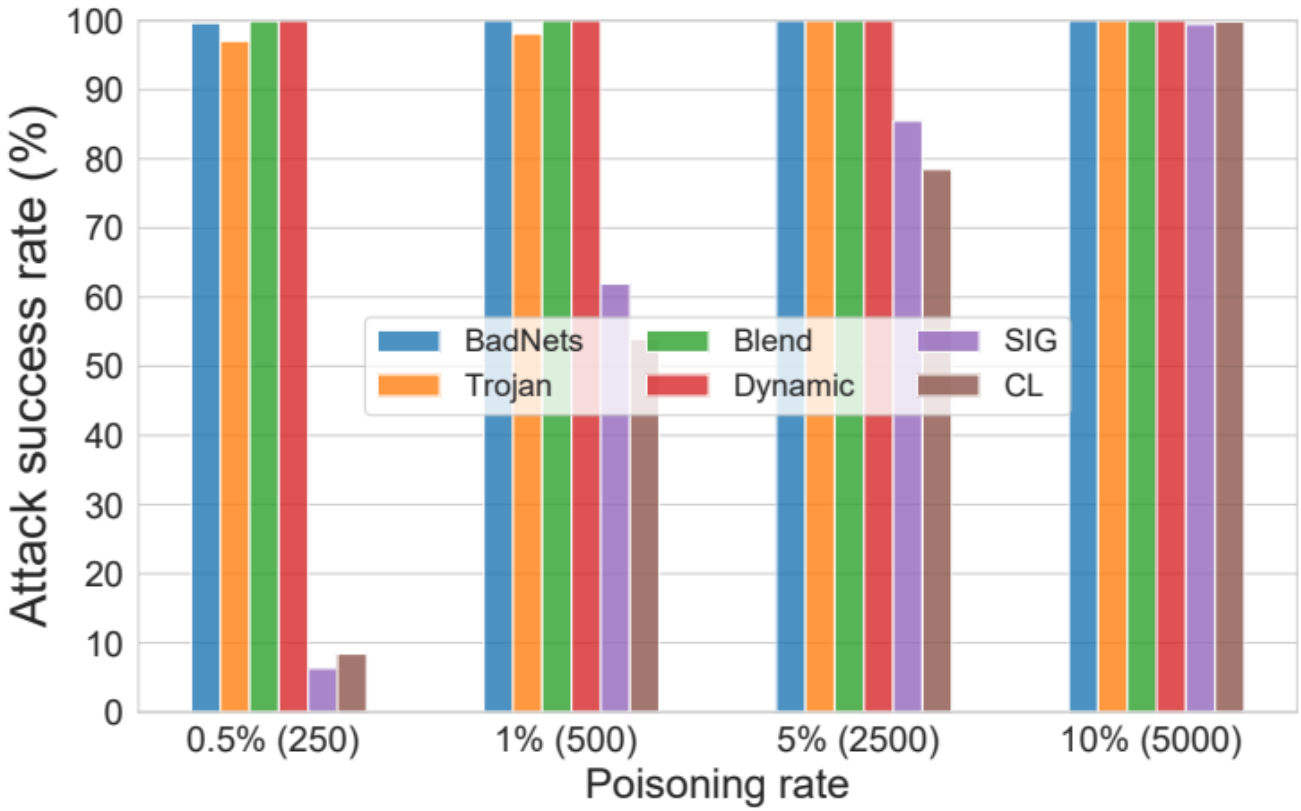
Overview

- Backdoor Attack:



Overview

- Performance of Backdoor Attacks:



Backdoor defenses

- ***Defense strategies*** against poisoning adversarial attacks were categorized in Gao et al. (2020) into:
  - Backdoor sample detection
  - Defense in training phase
  - Defense in testing phase

Backdoor sample detection

- Isolating backdoor examples at an early training stage can effectively prevent the backdoor attack
- Challenges:
  - The triggers can be imperceptible
  - Only a few samples (0.5%) and the proportion nor the distribution of the backdoored data is unkonwn in advance
  - The backdoor may have already been learned by the model even if the backdoor examples are identified at a later training stage

Backdoor defense

- ABL:

Anti-Backdoor Learning: Training Clean Models on Poisoned Data

- The defender’s goal:
- Training a model on the given dataset (potentially poisoned) that is as good as models trained on purely clean data.
- if an isolation method is used, the defender may identify only a subset of the backdoor examples

Backdoor defense

- ABL
- Probelm Formulation:

$$\mathcal{L} = \mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}}[\ell(f_\theta(\boldsymbol{x}), y)] = \underbrace{\mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}_c}[\ell(f_\theta(\boldsymbol{x}), y)]}_{\text{clean task}} + \underbrace{\mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}_b}[\ell(f_\theta(\boldsymbol{x}), y)]}_{\text{backdoor task}},$$

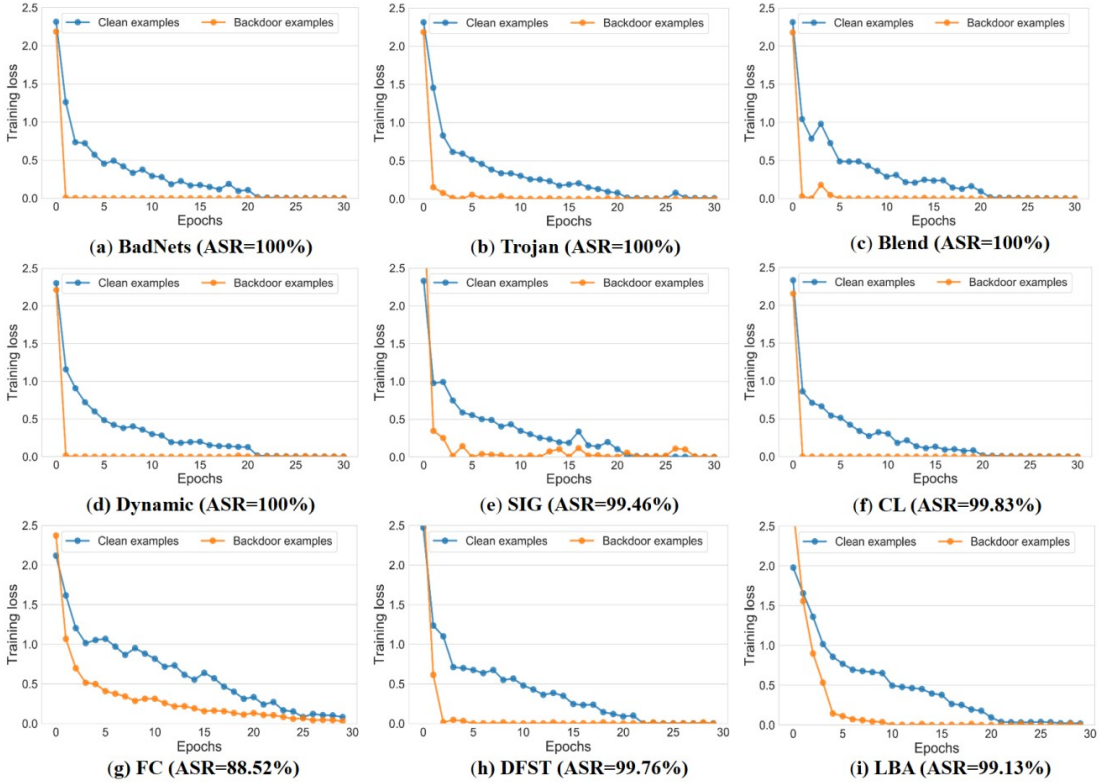
- The overall learning task is decomposed into two tasks where the first clean task is defined on the clean data Dc while the second backdoor task is defined on the backdoor data Db.

Backdoor defense

- ABL
- Weakness of backdoor attack:
- 1) the models learn backdoored data much faster than learning with clean data, and the stronger the attack the faster the model converges on backdoored data;
- 2) the backdoor task is tied to a specific class (the backdoor target class).

Backdoor defense

- ABL
- Distinctive Learning Behaviors on Backdoor Examples:



Backdoor defense

- ABL
- Backdoor Examples Isolation:

$$\mathcal{L}_{\text{ABL}}^{t} = \begin{cases} \mathcal{L}_{\text{LGA}} = \mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}}\left[\text{sign}(\ell(f_\theta(\boldsymbol{x}),y)-\gamma)\cdot\ell(f_\theta(\boldsymbol{x}),y)\right] & \text{if } 0 \le t < T_{te} \\ \mathcal{L}_{\text{GGA}} = \mathbb{E}_{(\boldsymbol{x},y)\sim\widehat{\mathcal{D}}_c}\left[\ell(f_\theta(\boldsymbol{x}),y)\right] - \mathbb{E}_{(\boldsymbol{x},y)\sim\widehat{\mathcal{D}}_b}\left[\ell(f_\theta(\boldsymbol{x}),y)\right] & \text{if } T_{te} \le t < T, \end{cases} \tag{3}$$

- The training process is composed of two stages: 1) backdoor isolation during early training, and 2) backdoor unlearning during later training.

Backdoor defense

- ABL
- Results:

Dataset	Types	No Defense		FP		MCR		NAD		**ABL (Ours)**	
		ASR	CA	ASR	CA	ASR	CA	ASR	CA	ASR	CA
CIFAR-10	*None*	0%	89.12%	0%	85.14%	0%	87.49%	0%	88.18%	0%	**88.41%**
	BadNets	100%	85.43%	99.98%	82.14%	3.32%	78.49%	3.56%	82.18%	**3.04%**	**86.11%**
	Trojan	100%	82.14%	66.93%	80.17%	23.88%	76.47%	18.16%	80.23%	**3.81%**	**87.46%**
	Blend	100%	84.51%	85.62%	81.33%	31.85%	76.53%	**4.56%**	82.04%	16.23%	**84.06%**
	Dynamic	100%	83.88%	87.18%	80.37%	26.86%	70.36%	22.50%	74.95%	**18.46%**	**85.34%**
	SIG	99.46%	84.16%	76.32%	81.12%	0.14%	78.65%	1.92%	82.01%	**0.09%**	**88.27%**
	CL	99.83%	83.43%	54.95%	81.53%	19.86%	77.36%	16.11%	80.73%	**0%**	**89.03%**
	FC	88.52%	83.32%	69.89%	80.51%	44.43%	77.57%	58.68%	81.23%	**0.08%**	**82.36%**
	DFST	99.76%	82.50%	78.11%	80.23%	39.22%	75.34%	35.21%	78.40%	**5.33%**	**79.78%**
	LBA	99.13%	81.37%	54.43%	79.67%	15.52%	78.51%	10.16%	79.52%	**0.06%**	**80.52%**
	CBA	90.63%	84.72%	77.33%	79.15%	38.76%	76.36%	33.11%	82.40%	**29.81%**	**84.66%**
	Average	97.73%	83.55%	75.07%	80.62%	24.38%	76.56%	20.40%	80.37%	**7.69%**	**84.76%**
GTSRB	*None*	0%	97.87%	0%	90.14%	0%	95.49%	0%	95.18%	0%	**96.41%**
	BadNets	100%	97.38%	99.57%	88.61%	1.00%	93.45%	0.19%	89.52%	**0.03%**	**96.01%**
	Trojan	99.80%	96.27%	93.54%	84.22%	2.76%	92.98%	0.37%	90.02%	**0.36%**	**94.95%**
	Blend	100%	95.97%	99.50%	86.67%	**6.83%**	92.91%	8.10%	89.37%	24.59%	**93.14%**
	Dynamic	100%	97.27%	99.84%	88.38%	64.82%	43.91%	68.71%	76.93%	**6.24%**	**95.80%**
	SIG	97.13%	97.13%	79.28%	90.50%	33.98%	91.83%	**4.64%**	89.36%	5.13%	**96.33%**
	Average	99.38%	96.80%	94.35%	87.68%	21.88%	83.01%	19.17%	87.04%	**7.27%**	**95.25%**
ImageNet Subset	*None*	0%	89.93%	0%	83.14%	0%	85.49%	0%	88.18%	0%	**88.31%**
	BadNets	100%	84.41%	97.70%	82.81%	28.59%	78.52%	6.32%	81.26%	**0.94%**	**87.76%**
	Trojan	100%	85.56%	96.39%	80.34%	6.67%	76.87%	15.48%	80.52%	**1.47%**	**88.19%**
	Blend	99.93%	86.15%	99.34%	81.33%	**19.23%**	75.83%	26.47%	82.39%	21.42%	**85.12%**
	SIG	98.60%	86.02%	78.82%	85.72%	25.14%	78.87%	5.15%	83.01%	**0.18%**	**86.42%**
	Average	99.63%	85.53%	93.06%	82.55%	19.91%	77.52%	13.35%	81.80%	**6.00%**	**86.87%**

Backdoor defense

- ABL
- Results:

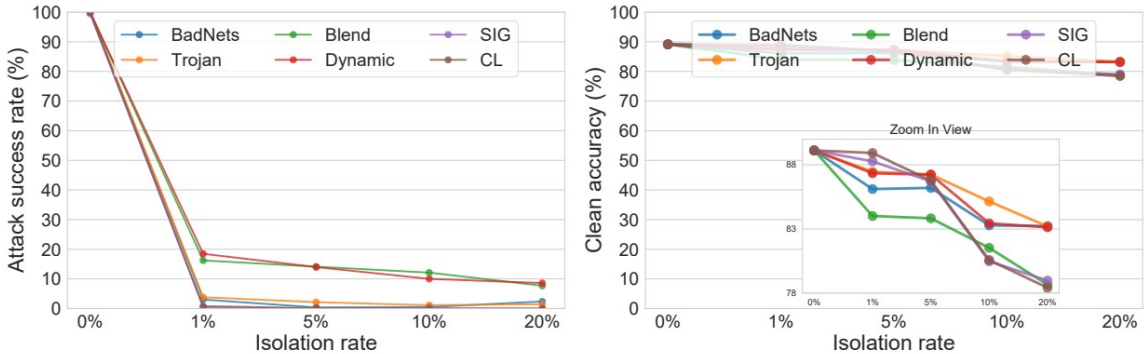


Figure 3: Performance of our ABL with different isolation rate $p \in [0.01, 0.2]$ on CIFAR-10 dataset. Left: attack success rate (ASR); Right: clean accuracy of ABL against 6 classic backdoor attacks.

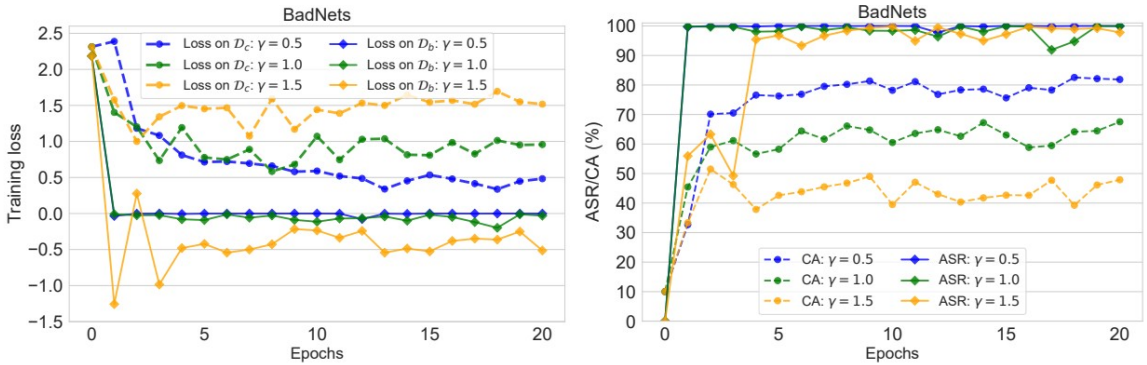


Figure 4: Separation effect of local gradient ascent with different γ on CIFAR-10 against BadNets. Left: Training loss on the ground truth backdoor ($\mathcal{D}_b$) and clean ($\mathcal{D}_c$) subsets; Right: Attack success rate (ASR) and clean accuracy (CA). The gap between the two lines of the same color becomes wider for larger γ, i.e., better separation effect.

Backdoor defense

- AIBD

Adversarial-Inspired Backdoor Defense via Bridging Backdoor and Adversarial Attacks

Jia-Li Yin[1, 2], Weijian Wang[1, 2], Lyhwa[2*], Wei Lin[3, 4*], Ximeng Liu[2, 5]
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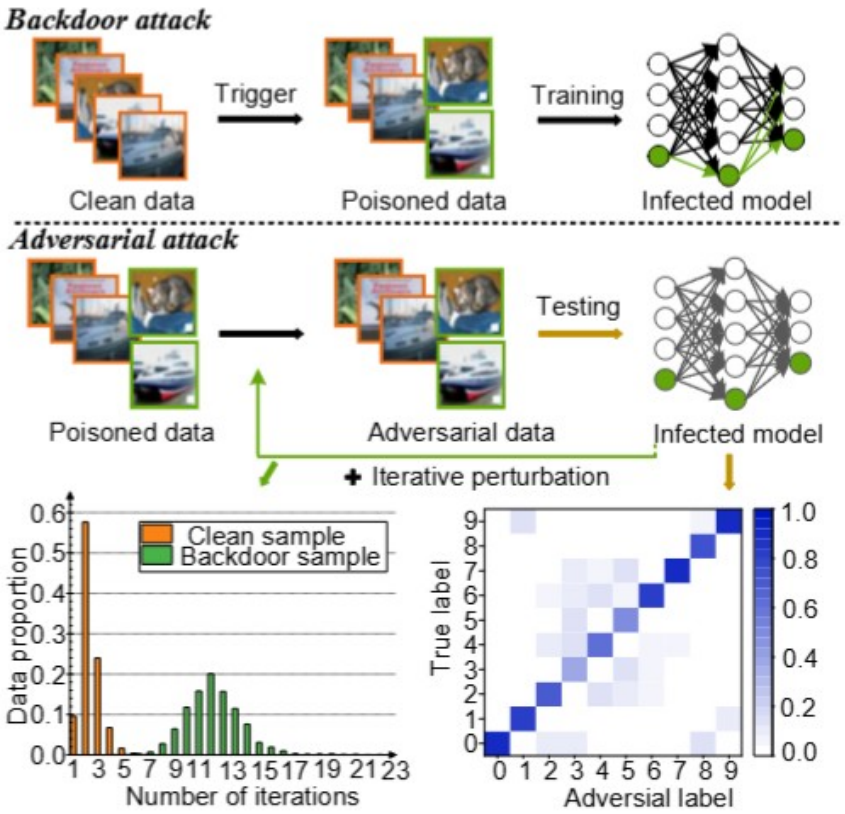
- While previous works generally treat these two attacks separately, the inherent connection between these two attacks is rarely explored.

Backdoor defense

- AIBD
- Motivation
- 1) the sample is harder to be turned into an AE when the trigger is presented; 2) the AE generated from backdoor samples are highly likely to be predicted as its true labels.



Backdoor defense

- AIBD

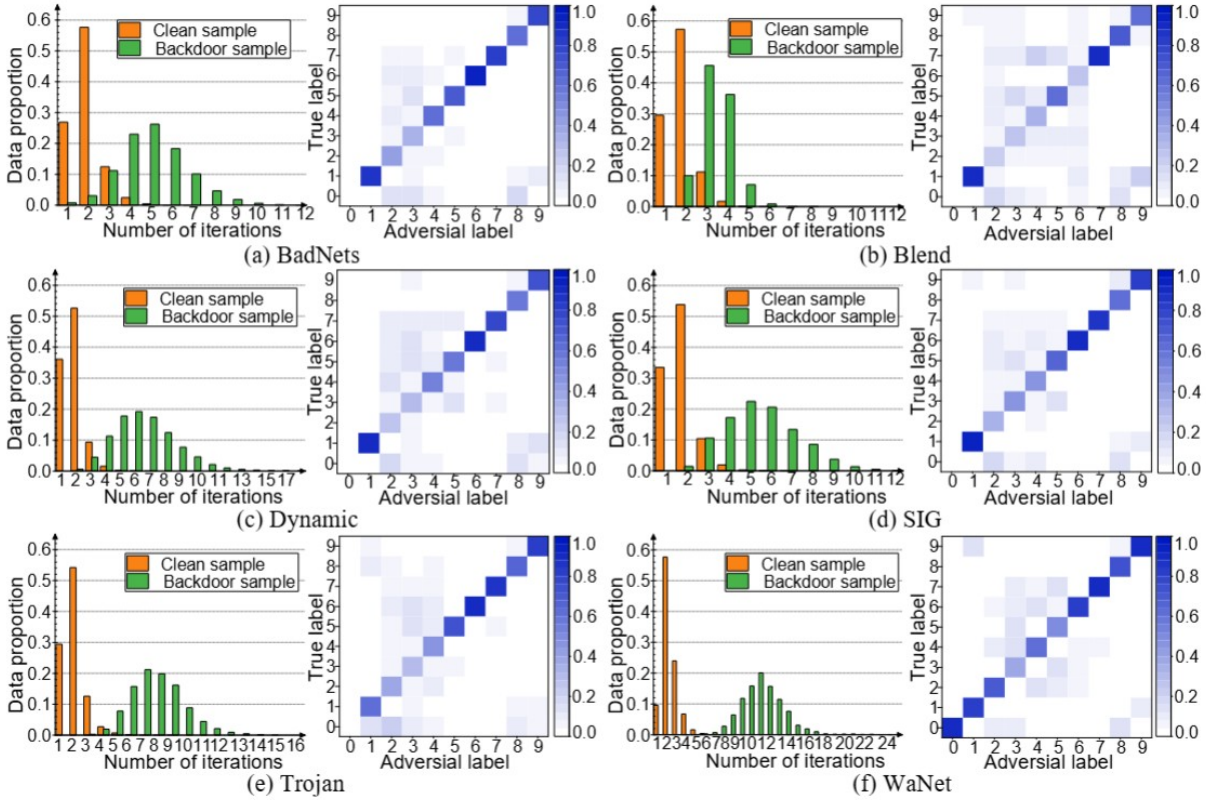


Figure 2: The performance of AEs generation from clean versus backdoor samples on injected models attacked by BadNet, Blend, Dynamic, SIG, Trojan, and WaNet. The experiment is conducted with ResNet-18 on the CIFAR-10 dataset under a poisoning rate of 10%, and the PGD attack is used to generate AEs in an iterative manner.

Backdoor defense

- AIBD
- Backdoor Sample Identification

$$\hat{\boldsymbol{x}}_{k+1} = \Pi_{\boldsymbol{x}+\mathcal{B}}(\hat{\boldsymbol{x}}_k + \alpha\,\mathrm{sign}(\nabla_{\boldsymbol{x}}\mathcal{L}(f(\boldsymbol{x};\theta), y)),$$

$$\begin{cases} \mathcal{D}^t \leftarrow (\boldsymbol{x}_i, y_i), Rank(k_i) \leq q, \\ \mathcal{D}^c \leftarrow (\boldsymbol{x}_i, y_i), otherwise. \end{cases}$$

- Adversarial Label Correction

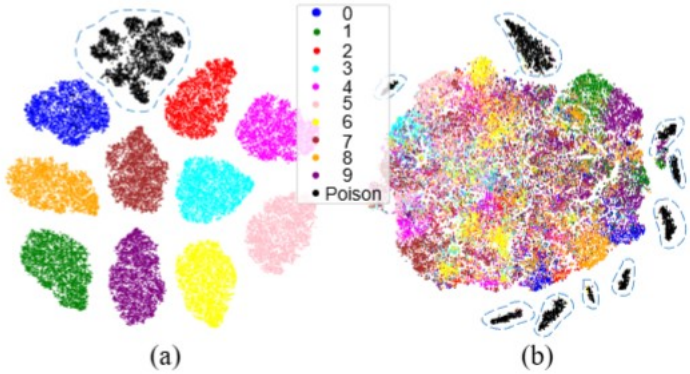
$$\hat{y}_i = f(\hat{\boldsymbol{x}}_i; \theta_p), \qquad \text{s.t. } \forall \boldsymbol{x}_i \in \mathcal{D}^t.$$

Backdoor defense

- AIBD
- Results

Types	BadNets		Blend		SIG		Dynamic		Trojan		Average	
Defense	TPR	FPR	TPR	FPR	TPR	FPR	TPR	FPR	TPR	FPR	TPR	FPR
None (Neural activation)	99.80	18.75	78.84	33.28	21.08	36.38	99.84	**0.01**	100.00	**0.74**	79.91	17.83
ABL (Loss isolation)	94.76	**0.68**	87.68	**0.64**	90.60	0.49	93.00	1.42	88.44	2.18	90.90	**1.08**
DBD (SCELoss)	80.67	48.52	**99.44**	47.39	38.36	50.61	21.48	51.50	**99.64**	47.38	67.92	49.08
SCALE-UP (Scale consistency)	32.36	32.54	39.36	40.05	27.64	27.38	28.72	27.79	30.56	30.98	31.73	31.75
AIBD (Ours)	**96.66**	1.40	97.26	1.12	**98.70**	1.70	**99.62**	1.22	99.49	1.30	**98.35**	1.35

Types	Unlearning		Random Labels		Adv. Labels	
	ACC	ASR	ACC	ASR	ACC	ASR
BadNets	84.82	0.72	82.36	1.84	**85.34**	**0.28**
Blend	67.69	**0.69**	83.32	4.22	**84.91**	0.73
SIG	75.58	**0.13**	81.63	0.27	**83.90**	0.33
WaNet	75.00	12.51	84.32	0.24	**85.39**	**0.28**
Trojan	84.06	0.37	84.84	0.28	**85.48**	**0.11**
Dynamic	83.97	2.20	82.98	0.69	**84.89**	**0.23**
Average	78.52	2.77	83.24	1.26	**84.99**	**0.33**



Backdoor defense

- AIBD
- Results

Dataset	Types	No Defense		ABL		NAD		DBD		CBD		**AIBD (Ours)**	
		ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR
CIFAR-10	*None*	88.28	0.00	66.84	0.00	86.21	0.00	78.53	0.00	79.86	0.00	**87.26**	0.00
	BadNets	87.18	98.43	82.68	6.94	81.81	1.68	82.94	0.80	84.01	4.29	**85.34**	**0.28**
	Blend	87.04	100.00	78.83	5.88	83.23	0.81	70.91	99.98	82.74	64.53	**84.91**	**0.73**
	SIG	85.75	99.98	81.43	3.90	83.18	1.20	83.52	0.44	80.34	83.31	**83.90**	**0.33**
	WaNet	86.35	100.00	79.22	17.19	82.71	1.77	84.26	6.61	85.13	3.14	**85.39**	**0.28**
	Trojan	87.48	100.00	80.95	1.38	84.57	3.59	81.22	2.24	80.73	10.88	**85.48**	**0.11**
	Dynamic	87.08	99.99	82.60	13.10	83.28	11.68	83.42	0.86	84.13	30.63	**84.89**	**0.23**
	Average	87.02	99.73	78.93	8.07	83.57	3.46	80.69	18.49	82.42	64.53	**85.31**	**0.33**
GTSRB	*None*	96.18	0.00	72.34	0.00	92.33	0.00	88.53	0.00	89.12	0.00	**96.40**	0.00
	BadNets	95.85	98.27	95.04	0.04	83.53	0.13	96.05	0.24	96.21	0.16	**95.84**	**0.00**
	Blend	95.02	99.97	92.19	4.69	80.46	**0.16**	93.72	6.36	94.16	0.90	**95.24**	0.19
	SIG	95.98	99.90	79.49	3.78	85.48	0.02	94.58	4.70	94.37	5.41	**95.34**	**0.00**
	WaNet	95.82	100.00	79.42	17.76	83.91	2.41	94.71	3.47	95.64	3.13	**95.62**	**0.00**
	Trojan	96.03	99.76	95.63	**0.01**	87.72	0.04	94.69	0.56	95.29	0.12	**96.25**	0.20
	Dynamic	96.37	99.83	95.11	1.79	90.51	0.20	95.86	5.16	**96.02**	0.96	95.86	**0.07**
	Average	95.89	99.62	87.03	4.68	86.28	0.50	94.02	3.42	94.40	1.78	**95.79**	**0.08**

Backdoor defense

- AIBD
- Results

Dataset	Types	No Defense		ABL		NAD		DBD		CBD		AIBD (Ours)	
		ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR
CIFAR-10	*None*	88.28	0.00	66.84	0.00	86.21	0.00	78.53	0.00	79.86	0.00	**87.26**	0.00
	BadNets	87.18	98.43	82.68	6.94	81.81	1.68	82.94	0.80	84.01	4.29	**85.34**	**0.28**
	Blend	87.04	100.00	78.83	5.88	83.23	0.81	70.91	99.98	82.74	64.53	**84.91**	**0.73**
	SIG	85.75	99.98	81.43	3.90	83.18	1.20	83.52	0.44	80.34	83.31	**83.90**	**0.33**
	WaNet	86.35	100.00	79.22	17.19	82.71	1.77	84.26	6.61	85.13	3.14	**85.39**	**0.28**
	Trojan	87.48	100.00	80.95	1.38	84.57	3.59	81.22	2.24	80.73	10.88	**85.48**	**0.11**
	Dynamic	87.08	99.99	82.60	13.10	83.28	11.68	83.42	0.86	84.13	30.63	**84.89**	**0.23**
	Average	87.02	99.73	78.93	8.07	83.57	3.46	80.69	18.49	82.42	64.53	**85.31**	**0.33**
GTSRB	*None*	96.18	0.00	72.34	0.00	92.33	0.00	88.53	0.00	89.12	0.00	**96.40**	0.00
	BadNets	95.85	98.27	95.04	0.04	83.53	0.13	96.05	0.24	96.21	0.16	**95.84**	**0.00**
	Blend	95.02	99.97	92.19	4.69	80.46	**0.16**	93.72	6.36	94.16	0.90	**95.24**	0.19
	SIG	95.98	99.90	79.49	3.78	85.48	0.02	94.58	4.70	94.37	5.41	**95.34**	**0.00**
	WaNet	95.82	100.00	79.42	17.76	83.91	2.41	94.71	3.47	95.64	3.13	**95.62**	**0.00**
	Trojan	96.03	99.76	95.63	**0.01**	87.72	0.04	94.69	0.56	95.29	0.12	**96.25**	0.20
	Dynamic	96.37	99.83	95.11	1.79	90.51	0.20	95.86	5.16	**96.02**	0.96	95.86	**0.07**
	Average	95.89	99.62	87.03	4.68	86.28	0.50	94.02	3.42	94.40	1.78	**95.79**	**0.08**

Backdoor defense

- Neural Cleanse

Neural Cleanse: Identifying and Mitigating Backdoor Attacks in Neural Networks

Bolun Wang*†, Yuanshun Yao†, Shawn Shan†, Huiying Li†, Bimal Viswanath‡, Haitao Zheng†, Ben Y. Zhao†
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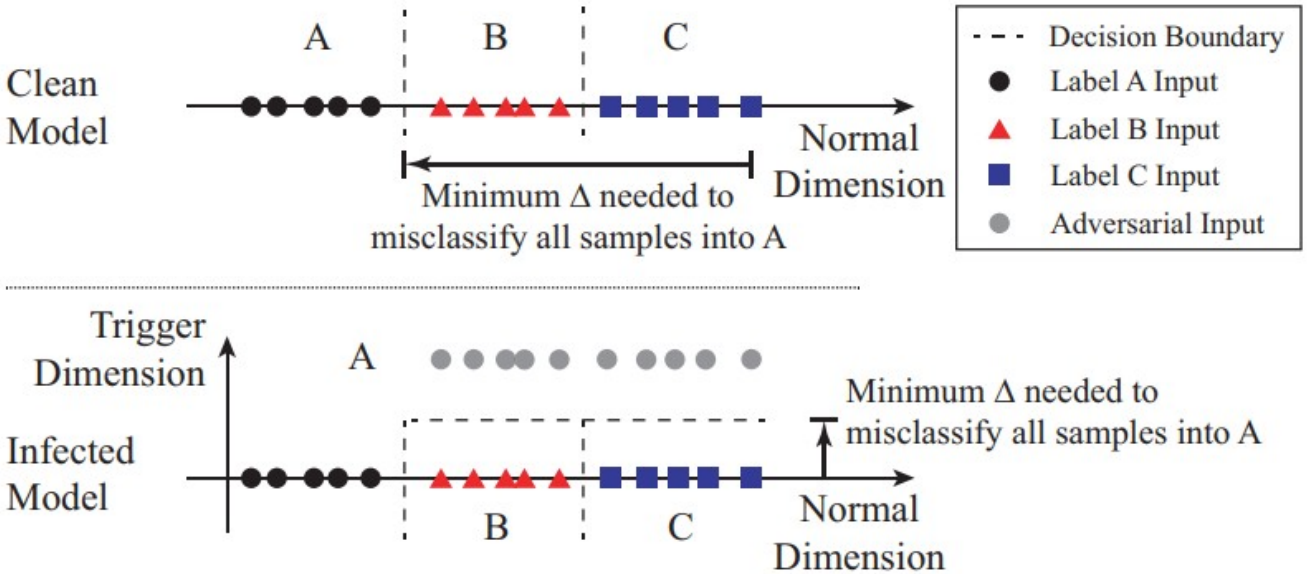
*bolunwang@cs.ucsb.edu, †{ysyao, shansixiong, huiyingli, htzheng, ravenben}@cs.uchicago.edu, ‡vbimal@cs.vt.edu

- i) an early filter for adversarial inputs that identifies inputs with a known trigger;
- ii) a model patching algorithm based on neuron pruning;
- iii) a model patching algorithm based on unlearning.

Backdoor defense

- Neural Cleanse
- Detection



Backdoor defense

- Neural Cleanse
- Detection

$$\delta_{\forall \to t} \leq |T_t| << \min_{i, i \neq t} \delta_{\forall \to i}$$

- **Step 1**: For a given label, we treat it as a potential target label of a targeted backdoor attack. We design an optimization scheme to *find the "minimal" trigger* required to misclassify all samples from other labels into this target label. In the vision domain, this trigger defines the smallest collection of pixels and its associated color intensities to cause misclassification.
- **Step 2**: We repeat Step 1 for each output label in the model. For a model with $N = |\mathbb{L}|$ labels, this produces N potential "triggers".
- **Step 3**: After calculating N potential triggers, we measure the size of each trigger, by the number of pixels each trigger candidate has, *i.e.* how many pixels the trigger is replacing. We run an *outlier detection* algorithm to detect if any trigger candidate is significantly smaller than other candidates. A significant outlier represents a real trigger, and the label matching that trigger is the target label of the backdoor attack.

Backdoor defense

- Neural Cleanse
- Identification

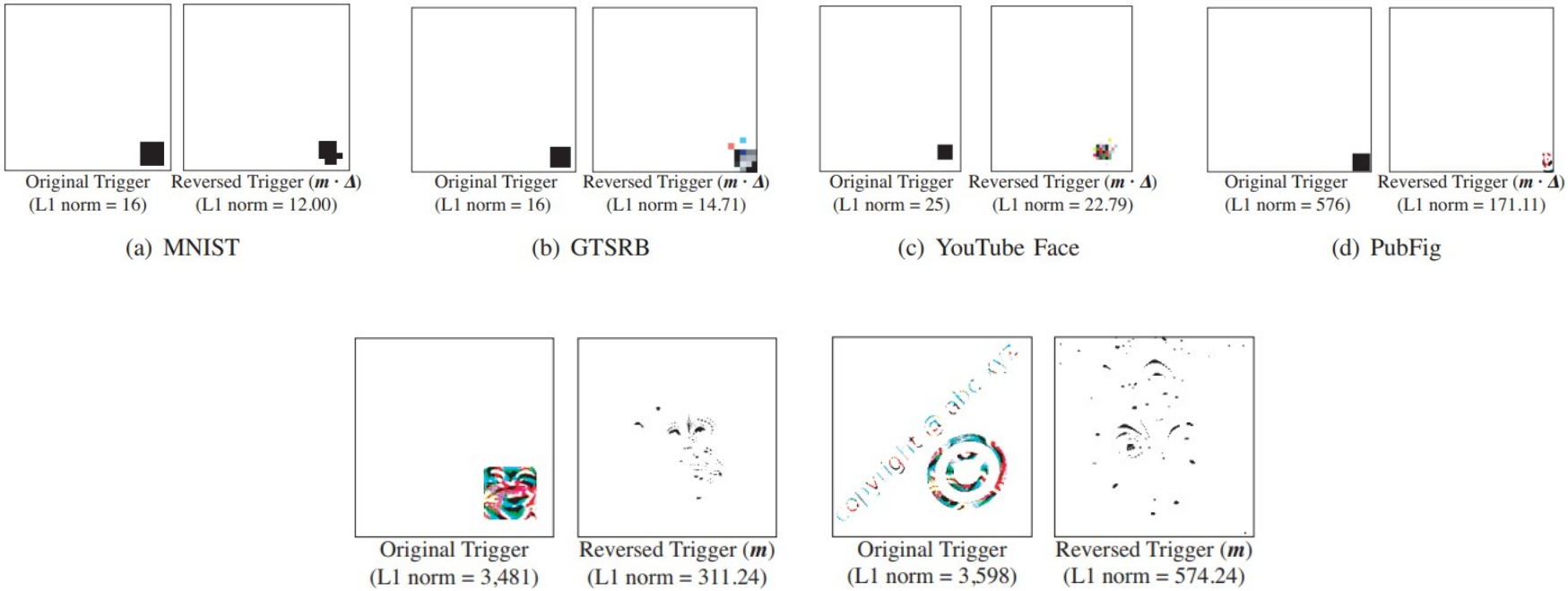
$$A(\boldsymbol{x}, \boldsymbol{m}, \boldsymbol{\Delta}) = \boldsymbol{x}'$$
$$\boldsymbol{x}'_{i,j,c} = (1 - \boldsymbol{m}_{i,j}) \cdot \boldsymbol{x}_{i,j,c} + \boldsymbol{m}_{i,j} \cdot \boldsymbol{\Delta}_{i,j,c}$$

- Reverse Engineering Triggers

$$\min_{\boldsymbol{m}, \boldsymbol{\Delta}} \quad \ell(y_t, f(A(\boldsymbol{x}, \boldsymbol{m}, \boldsymbol{\Delta}))) + \lambda \cdot |\boldsymbol{m}|$$
$$\text{for} \quad \boldsymbol{x} \in \boldsymbol{X}$$

Backdoor defense

- Neural Cleanse
- Identification



Backdoor defense

- Neural Cleanse
- Mitigation
  - 1-Filter for Detecting Adversarial Inputs
  - 2-Patching DNN via Neuron Pruning
  - 3-Patching DNNs via Unlearning

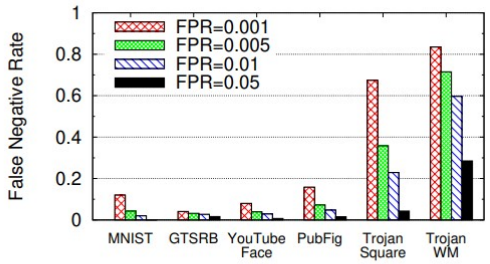


Fig. 8. False negative rate of proactive adversarial image detection when achieving different false positive rate.

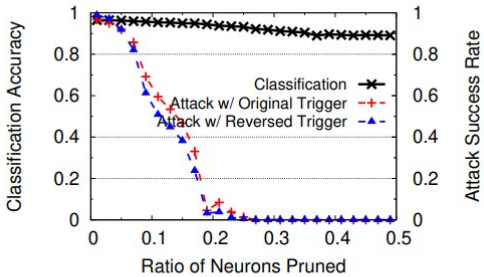


Fig. 9. Classification accuracy and attack success rate when pruning trigger-related neurons in GTSRB (traffic sign recognition w/ 43 labels).

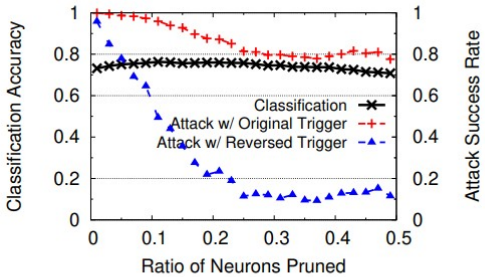


Fig. 10. Classification accuracy and attack success rate when pruning trigger-related neurons in Trojan Square (face recognition w/ 2,622 labels).